TABLE OF CONTENTS

Page

4. NOISE IMPACT 4-1

4.1 Introduction 4-1
4.2 Environmental Legislation, Standards and Guidelines 4-1
4.3 Description of the Environment 4-5
4.4 Identification of Noise Sensitive Receivers 4-5
4.5 Identification of Environmental Impacts 4-12
4.6 Assessment Methodology 4-15
4.7 Prediction and Evaluation of Environmental Impacts 4-17
4.8 Mitigation of Environmental Impacts 4-25
4.9 Evaluation of Residual Impact 4-39
4.10 Environmental Monitoring and Audit 4-39
4.11 Conclusion 4-39

List of Tables

Table 4.1 Noise Standards for Daytime Construction Activities
Table 4.2 Area Sensitivity Ratings (ASRs)
Table 4.3 Acceptable Noise Levels under the Technical Memorandum on Noise from Construction Work other than Percussive Piling
Table 4.4 Noise Standards for Road Traffic Noise
Table 4.5 Noise Criteria for Fixed Noise Sources
Table 4.6 Representative Noise Sensitive Receivers (NSRs) and Selected Noise Assessment Points (NAPs) for Construction Phase
Table 4.7 Representative Noise Sensitive Receivers (NSRs) and Selected Noise Assessment Points (NAPs) for Traffic Noise Assessment
Table 4.8 Representative Noise Sensitive Receivers (NSRs) and Selected Noise Assessment Points (NAPs) for Fixed Noise Assessment
Table 4.9 Summary of Prevailing Background Noise Measurement
Table 4.10 Area Sensitivity Rating and Noise Criteria for Fixed Noise Impact Assessment
Table 4.11 List of Potential Concurrent Projects
Table 4.12 Fixed Noise Source from the Project
Table 4.13 Plant Inventory for Key Construction Activities
Table 4.14 Summary of Predicted Road Traffic Noise Assessment Results under Unmitigated Scenario (Year 2046)
Table 4.15 Fixed Plant Inventory
Table 4.16 Quieter Construction Method / Equipment
Table 4.17 Mitigation Measures for Road Traffic Noise Impact
Table 4.18 Summary of Predicted Road Traffic Noise Assessment Results under the Mitigated Scenario (with implementation of Acoustic Window / Acoustic Balconies and Low Noise Road Surfacing)
Table 4.19 Estimated Number of Existing Residential Dwellings/ Rooms Benefited and Protected by Mitigation Measures

List of Figures

60722948/A10/FIGURE 4.1	300m Assessment Area for Noise Assessment
60722948/A10/FIGURE 4.2	Locations of Representative Noise Sensitive Receivers for Construction Noise Impact Assessment (Key Plan)
60722948/A10/FIGURE 4.2a	Locations of Representative Noise Sensitive Receivers for Construction Noise Impact Assessment (Sheet 1 of 4)
60722948/A10/FIGURE 4.2b	Locations of Representative Noise Sensitive Receivers for Construction Noise Impact Assessment (Sheet 2 of 4)
60722948/A10/FIGURE 4.2c	Locations of Representative Noise Sensitive Receivers for Construction Noise Impact Assessment (Sheet 3 of 4)
60722948/A10/FIGURE 4.2d	Locations of Representative Noise Sensitive Receivers for Construction Noise Impact Assessment (Sheet 4 of 4)
60722948/A10/FIGURE 4.3	Locations of Representative Noise Sensitive Receivers for Road Traffic Noise Impact Assessment (Key Plan)
60722948/A10/FIGURE 4.3a	Locations of Representative Noise Sensitive Receivers for Road Traffic Noise Impact Assessment (Sheet 1 of 4)
60722948/A10/FIGURE 4.3b	Locations of Representative Noise Sensitive Receivers for Road Traffic Noise Impact Assessment (Sheet 2 of 4)
60722948/A10/FIGURE 4.3c	Locations of Representative Noise Sensitive Receivers for Road Traffic Noise Impact Assessment (Sheet 3 of 4)
60722948/A10/FIGURE 4.3d	Locations of Representative Noise Sensitive Receivers for Road Traffic Noise Impact Assessment (Sheet 4 of 4)
60722948/A10/FIGURE 4.4	Prevailing Noise Measurement Location
60722948/A10/FIGURE 4.5	Extent of Project Road
60722948/A10/FIGURE 4.6	Location of Fixed Noise Source and Representative Noise Sensitive Receivers for Fixed Noise Impact Assessment
60722948/A10/FIGURE 4.7a	Location of Noise Mitigation Measures
60722948/A10/FIGURE 4.7b	Location of Noise Mitigation Measures

List of Appendices

Appendix 4.1	Photographs of the Existing Representative Noise Sensitive Receivers
Appendix 4.2	2046 Peak Hour Traffic Forecasts and the Road Vertical Profile
Appendix 4.3	Endorsement of the Traffic Forecast Data from Transport Department
Appendix 4.4	Detailed Results of Road Traffic Noise Assessment (Unmitigated Scenario)
Appendix 4.5	Detailed Results of Road Traffic Noise Assessment (Mitigated Scenario)
Appendix 4.6	Agreement from Project Proponent of MTL Project

4. NOISE IMPACT

4.1 Introduction

4.1.1 This section presents the assessment of potential noise impacts associated with the construction and operation of the Project. The noise impact assessment has been conducted in accordance with the requirement of Annex 5 and Annex 13 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) as well as the requirements set out in Clause 3.5.4 of the Environmental Impact Assessment (EIA) Study Brief (No. ESB-375/2025).

4.2 Environmental Legislation, Standards and Guidelines

4.2.1 Noise impact has been assessed in accordance with the criteria and methodology given in the Technical Memoranda (TM) under the Noise Control Ordinance (NCO), and the EIAO-TM, Guidance Notes under EIAO (EIAO-GN).

4.2.2 Assessment procedures and standards are set out in the following TMs and Guidance Notes:-

- Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM);
- Technical Memorandum for the Assessment of Noise from Places other than Domestic Premises, Public Places or Construction Sites (IND-TM);
- Technical Memorandum on Noise from Construction Work other than Percussive Piling (GW-TM);
- Technical Memorandum on Noise from Percussive Piling (PP-TM);
- EIAO-GN 9/2023 Preparation of Construction Noise Impact Assessment under the Environmental Impact Assessment Ordinance (EIAO);
- EIAO-GN 16/2023 Preparation of Fixed Noise Sources Impact Assessment Under the EIAO; and
- EIAO-GN 12/2023 Road Traffic Noise Impact Assessment under the EIAO.

Construction Noise Criteria

General Construction Works

4.2.3 The NCO provides the statutory framework for noise control of construction works, other than percussive piling, using powered mechanical equipment (PME) between the hours of 1900 and 0700 hours or at any time on Sundays and general holiday (i.e. restricted hours). Noise from construction activities taking place at 0700 to 1900 hours on any day not being a Sunday or general holiday is subject to the Noise Standards for Daytime Construction Activities stated in Table 1B of Annex 5 in the EIAO-TM. The criteria are summarised in **Table 4.1**.

Table 4.1 Noise Standards for Daytime Construction Activities

Uses	0700 to 1900 Hours on Any Day Not Being a Sunday or General Holiday $L_{eq\ (30mins)}$ dB(A)
• All domestic premises, • Temporary housing accommodation; • Hostels; • Convalescent homes; and • Homes for the aged	75
• Places of public worship • Courts of law, and • Hospitals and medical clinics	70
• Educational institutions (including kindergartens and nurseries)	70 65 (during examination)

Note:
(1) The above standards apply to uses which reply on opened windows for ventilation and are assessed at 1m from the external façade.

General Construction Works during Restricted Hours

4.2.4 Regarding the assessment on construction noise impact during restricted hours (i.e., between 19:00 and 07:00 hours and at any time on Sundays and general holidays), the NCO designates Acceptable Noise Levels (ANLs) for Noise Sensitive Receivers (NSRs) based on an Area Sensitivity Rating (ASR), which is determined by the characteristics of the area within which they are located as described in **Table 4.2**. Within these areas, the presence of "influencing factors" (such as the presence of industrial area of major roads) can further affect the ASR and hence the acceptable noise levels.

Table 4.2 Area Sensitivity Ratings (ASRs)

Type of Area Containing NSR	Degree to which NSR is affected by Influencing Factor		
	Not Affected	Indirectly Affected	Directly Affected
Rural area, including country parks or village type developments	A	B	B
Low density residential area consisting of low-rise or isolated high-rise developments	A	B	C
Urban area	B	C	C

Type of Area Containing NSR	Degree to which NSR is affected by Influencing Factor		
	Not Affected	Indirectly Affected	Directly Affected
Area other than those above	B	B	C

Notes:
(1) "Country park" means an area that is designated as a country park pursuant to section 14 of the Country Parks Ordinance;
(2) Directly affected" means that the NSR is at such a location that noise generated by the IF is readily noticeable at the NSR and is a dominant feature of the noise climate of the NSR;
(3) "Indirectly affected" means that the NSR is at such a location that noise generated by the IF, whilst noticeable at the NSR, is not a dominant feature of the noise climate of the NSR;
(4) "Not affected" means that the NSR is at such a location that noise generated by the IF is not noticeable at the NSR; and
(5) "Urban area" means an area of high density, diverse development including a mixture of such elements as industrial activities, major trade or commercial activities and residential premises.

4.2.5 On all days between 1900 and 0700 hours and at any time on Sundays and general holidays, the use of PME for the purpose of carrying out construction works is prohibited unless a Construction Noise Permit (CNP) has been obtained. The CNP may be granted provided that the ANL for the NSRs can be complied with. ANLs are assigned depending upon the ASR and are given in **Table 4.3**.

Table 4.3 Acceptable Noise Levels under the Technical Memorandum on Noise from Construction Work other than Percussive Piling

Time Period	Acceptable Noise Level (ANLs), dB(A)		
	ASR A	ASR B	ASR C
All days during the evening (1900 to 2300 hours), and general holidays (including Sundays) during the daytime and evening (0700 to 2300 hours)	60	65	70
All days during the night-time (2300 to 0700 hours)	45	50	55

4.2.6 The Noise Control Authority will consider a well-justified CNP application, for construction works within restricted hours as guided by the relevant TM issued under the NCO. The Noise Control Authority will consider contemporary conditions / situations of adjoining land uses and any previous complaints against construction activities at the site before making his decision in granting a CNP. Nothing in this EIA shall bind the Noise Control Authority in making his decision. If a CNP is to be issued, the Noise Control Authority shall include in it any condition he thinks fit. Failure to comply with any such conditions will lead to cancellation of the CNP and prosecution action under the NCO.

Percussive Piling Works

4.2.7 Percussive piling is prohibited between 1900 and 0700 hours on any weekday not being a general holiday and at any time on Sunday or general holiday. A CNP is required for the carrying out of percussive piling between 0700 and 1900 hours on any day not being a general holiday. PP-TM sets out the requirements for working and determination of the permitted hours of operations for the CNP applications. The

permitted hours of operations would be 3, 5 or 12 hours per day depending on the types of percussive piling and the predicted noise impact at NSRs.

Operation Noise

Road Traffic Noise

4.2.8 For road traffic noise, the following $L_{10(1\ hour)}$ criteria stipulated in Annex 5, Table 1A of EIAO-TM are adopted for different types of NSRs which rely on opened windows for ventilation. Relevant criteria are listed in **Table 4.4**.

Table 4.4 Noise Standards for Road Traffic Noise

Uses	Peak Hour Traffic Noise Level, $L_{10(1hr)}$ dB(A)
• All domestic premises, • Temporary housing accommodation; • Hostels; • Convalescent homes; and • Homes for the aged	70
• Educational institutions (including kindergartens and nurseries) • Places of public worship; and • Courts of law	65
• Hospitals and medical clinics	55

Note:
(1) The above standards apply to uses which reply on opened windows for ventilation and are assessed at 1m from the external façade.

Fixed Noise

4.2.9 Fixed noise sources are controlled by Section 13 of the NCO, IND-TM. For planned fixed noise sources, 5dB(A) below the appropriate ANLs as shown in Table 2 of the IND-TM or the prevailing background noise levels (for quiet areas with level 5 dB(A) below the ANL) as recommended in the Annex 5 of EIAO-TM for planning purposes are applied to this study. The ANLs and criteria for different ASRs are summarised in **Table 4.5** below.

Table 4.5 Noise Criteria for Fixed Noise Sources

Time Period	Noise Criteria for Different Area Sensitivity Rating ($L_{eq\ 30\ min}$, dB(A)) (1)(2)					
	ANL, dB(A)			ANL-5, dB(A) for Planned Fixed Noise Sources		
	ASR A	ASR B	ASR C	ASR A	ASR B	ASR C
Day (0700 to 1900 hrs)	60	65	70	55	60	65
Evening (1900 to 2300 hrs)	60	65	70	55	60	65
Night (2300 to 0700 hrs)	50	55	60	45	50	55

Notes:
(1) The noise criteria apply to uses which rely on opened window for ventilation only.
(2) The above standards should be viewed as the maximum permissible noise levels assessed at 1 m from the external façade.

4.2.10 In any event, the area sensitivity ratings (ASRs) adopted in this EIA Report is for indicative assessment only. Therefore, the Noise Control Authority shall determine noise impact from concerned fixed noise sources on the basis of prevailing legislation and practices being in force, and taking account of contemporary conditions / situations of adjoining land uses. Nothing in this EIA Report shall bind the Noise Control Authority in the context of law enforcement against any of the fixed noise sources being assessed.

4.3 Description of the Environment

4.3.1 The Project connects the Loop in the east and the Ma Tso Lung areas in the west. The surrounding environment consists of rural settlements, farmlands, and fishponds. The prevailing noise climate in the assessment area is dominated by the existing traffic noise from Ma Tso Lung Road, Liu Pok Road and the Border Road.

4.4 Identification of Noise Sensitive Receivers

4.4.1 In accordance with Annex 13 of the EIAO-TM, all domestic premises including temporary housing, hostels, educational institutions (including kindergartens and nurseries), hospitals, medical clinics, homes for the aged, convalescent homes, places of public worship, courts of law are considered as Noise Sensitive Receivers (NSRs).

4.4.2 The assessment area for construction noise, road traffic noise and fixed noise source for the Project is defined as 300m away from the Project boundary in accordance with Clauses 1.1.1 (a), 2.1.1(a) and 3.1.1(a), Appendix C of the EIA Study Brief and is presented in **60722948/A10/FIGURE 4.1**. The representative existing NSRs within the assessment area for noise impact assessments of the construction and operation phases have been identified based on the observations from the site visits dated 18 April 2024, 15 August 2024, 14 January 2025, and 11 September 2025. Some other

structures closer to the project boundary than the identified representative NSRs were inspected. They were of non-noise sensitive use, and thus not considered as representative NSRs.

4.4.3 According to the latest land use plan under "Agreement No. CE 46/2022 – Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Feasibility Study (hereinafter referred to as "MTL Project")", Site Res 3 and Site Res 6 are designated for residential use. During the course of the EIA Study, detailed building layout plans for the potential residential development at Site Res 3 and Site Res 6 were not yet available. Therefore, an assessment point has been conservatively selected at the boundary of these sites for evaluation purposes. Based on the latest available information, the construction of MTL Project is anticipated to commence in Year 2028 with population intake expected in Year 2034, which is after the completion of the Project (i.e. Year 2031). As such, the assessment points at Site Res 3 and Site Res 6 have been included for operation phase assessment only. The maximum proposed building height for Sites Res 3 and Res 6 is +155mPD. For the purpose of the assessment, it was assumed that the planned NSRs at Sites Res 3 and Res 6 will have up to 45 storeys. It is noted that the construction of Project will commence in Year 2027, while the MTL Project construction is tentatively scheduled for 2028 to 2034. As such, some existing NSRs which fall under the MTL Project area, may be affected by construction noise impacts from the Project. Accordingly, these existing NSRs have been identified for the construction phase assessment.

4.4.4 With reference to the latest Outline Zoning Plan (OZP) (Approved Lok Ma Chau Loop Outline Zone Plan No. S/LMCL/2 [1]), certain Other Specified Uses (OU) and Government, Institution or Community (G/IC) zonings are reserved for non-noise sensitive uses, including district cooling system, sewage treatment works and electricity substation. The northwest G/IC zoning is reserved for fire station-cum-ambulance depot, Hong Kong Police Force (HKPF) facilities and boundary crossing facilities. It is assumed that these facilities, along with the planned commercial uses will not rely on opened windows for ventilation and are therefore considered as noise tolerant. According to the Outline Zoning Plan (OZP), the "Other Specified Uses" (OU) zoning located on the southwestern side of the Project is designated for research and development, education, and cultural and creative industries. This zone is planned to accommodate student hostels, staff quarters, academic buildings, and administrative facilities. As stated in the approved EIA Report for the Development of Lok Ma Chau Loop (AEIAR-176/2013), central air conditioning is assumed to be provided for academic and administrative buildings. Therefore, only the student hostels and staff quarters are considered NSRs.

4.4.5 The locations of the identified representative NSRs and Noise Assessment Point (NAPs) for construction noise, road traffic noise and fixed noise impact assessments are presented in **60722948/A10/FIGURE 4.2, 4.3,** and **4.6**. A list of the representative NSRs and NAPs are presented in **Table 4.6** to **Table 4.8**. The photographs of the existing NSRs are presented in **Appendix 4.1**.

4.4.6 In order to evaluate the noise impacts during construction and operation phases associated with the Project, representative existing/planned NSRs located within the assessment area have been identified for assessment. Only the first layer of NSRs have been identified for assessment because it would provide acoustic shielding to those receivers at further distance behind.

[1] https://www.ozp.tpb.gov.hk/api/Plan/PlanNote?planNo=S%2fLMCL%2f2&lang=EN&ext=pdf&dType=in

Table 4.6 Representative Noise Sensitive Receivers (NSRs) and Selected Noise Assessment Points (NAPs) for Construction Phase

NSR ID	NSR Description	NAP ID	NAP Description	Land Use[(1)]	Number of Storey
Existing NSRs					
MTLRLPR	Village Houses near the junction of Ma Tso Lung Road and Liu Pok Road	MTLRLPR1	Temporary Structure to the East of Lamp Post N4001	R	1
		MTLRLPR2	Temporary Structure to the South of Lamp Post CD0749	R	1
		MTLRLPR3	Temporary Structure to the West of Lamp Post CD0983	R	1
MTLR	Village Houses near Ma Tso Lung Road	MTLR1	Temporary Structure to the Northeast of Lamp Post VA4082	R	1
		MTLR2	Village House to the North of Lamp Post VG3124	R	1
		MTLR3	Village House to the North of Lamp Post VA4080	R	2
		MTLR4	Temporary Structure to the North of Lamp Post VD9113	R	1
		MTLR5	Village House to the Northeast of Lamp Post VD9113	R	2
MTLST	Ma Tso Lung San Tsuen	MTLST1	Temporary Structure to the Northeast of Lamp Post VD3135	R	1
		MTLST2	Temporary Structure Adjacent to 100 Ma Tso Lung San Tsuen	R	2
		MTLST3	Temporary Structure to the North of Lamp Post VD3133	R	1
		MTLST4	Temporary Structure to the West of Lamp Post VD3131	R	1
		MTLST5	1 Ma Tso Lung San Tsuen	R	2
		MTLST6	3 Ma Tso Lung San Tsuen	R	2
		MTLST7	11 Ma Tso Lung San Tsuen	R	2
		MTLST8	17 Ma Tso Lung San Tsuen	R	2
MTLSYST	Ma Tso Lung Shun Yee San Tsuen	MTLSYST2	1 Ma Tso Lung Shun Yee San Tsuen Block A	R	1
		MTLSYST8	5 Ma Tso Lung Shun Yee San Tsuen Block D	R	1
		MTLSYST9	Village House to the East of Lamp Post N4004	R	2

NSR ID	NSR Description	NAP ID	NAP Description	Land Use(1)	Number of Storey
PH	Village Houses near Ping Hang	PH1	Temporary Structure to the South of Lamp Post VG3160	R	1
TKH	Village Houses near Tse Koo Hang	TKH1	Village House to the West of Lamp Post VG4766	R	2
		TKH2	Village House to the Southwest of Lamp Post VG4766	R	2
		TKH3	Village House in DD96 1427	R	2
		TKH4	Temporary Structure to the East of Lamp Post VG3141	R	1
MTLRS	Village Houses near the Southern Area of Ma Tso Lung Road	MTLRS1	Village House near to Lamp Post N3984	R	1
KTN	Village House at Kwu Tung North	KTN1	Temporary Structure to the West of Lamp Post VG5734	R	1

Note:
(1) Residential – R.

Table 4.7 Representative Noise Sensitive Receivers (NSRs) and Selected Noise Assessment Points (NAPs) for Traffic Noise Assessment

NSR ID	NSR Description	NAP ID	NAP Description	Land Use[(1)]	Number of Storey
Existing NSRs					
MTLRLPR	Village Houses near the junction of Ma Tso Lung Road and Liu Pok Road	MTLRLPR1	Temporary Structure to the East of Lamp Post N4001	R	1
		MTLRLPR2	Temporary Structure to the South of Lamp Post CD0749	R	1
		MTLRLPR3	Temporary Structure to the West of Lamp Post CD0983	R	1
MTLR	Village Houses near Ma Tso Lung Road	MTLR1	Temporary Structure to the Northeast of Lamp Post VA4082	R	1
		MTLR2	Village House to the North of Lamp Post VG3124	R	1
		MTLR3	Village House to the North of Lamp Post VA4080	R	2
		MTLR4	Temporary Structure to the North of Lamp Post VD9113	R	1
		MTLR5	Village House to the Northeast of Lamp Post VD9113	R	2
MTLST	Ma Tso Lung San Tsuen	MTLST3	Temporary Structure to the North of Lamp Post VD3133	R	1
MTLSYST	Ma Tso Lung Shun Yee San Tsuen	MTLSYST1	Village House to the Northeast of Lamp Post EC1009	R	2
		MTLSYST2	1 Ma Tso Lung Shun Yee San Tsuen Block A	R	1
		MTLSYST3	6 Ma Tso Lung Shun Yee San Tsuen Block A	R	1
		MTLSYST4	2 Ma Tso Lung Shun Yee San Tsuen Block B	R	1
		MTLSYST5	1 Ma Tso Lung Shun Yee San Tsuen Block C	R	1
		MTLSYST6	4A Ma Tso Lung Shun Yee San Tsuen Block C	R	2
		MTLSYST7	1 Ma Tso Lung Shun Yee San Tsuen Block D	R	1
		MTLSYST8	5 Ma Tso Lung Shun Yee San Tsuen Block D	R	1
		MTLSYST9	Village House to the East of Lamp Post N4004	R	2

NSR ID	NSR Description	NAP ID	NAP Description	Land Use[1]	Number of Storey
PH	Village Houses near Ping Hang	PH1	Temporary Structure to the South of Lamp Post VG3160	R	1
TKH	Village Houses near Tse Koo Hang	TKH1	Village House to the West of Lamp Post VG4766	R	2
		TKH2	Village House to the Southwest of Lamp Post VG4766	R	2
		TKH3	Village House in DD96 1427	R	2
		TKH4	Temporary Structure to the East of Lamp Post VG3141	R	1
MTLRS	Village Houses near the Southern Area of Ma Tso Lung Road	MTLRS1	Village House near to Lamp Post N3984	R	1
Planned NSRs					
TL	Planned Development within the Loop	TL1	Planned Student Hostel and Staff Quarter	R[3]	9
		TL2	Planned Student Hostel and Staff Quarter	R[3]	9
		TL3	Planned Student Hostel and Staff Quarter	R[3]	6
MTL_RES3[2]	Planned Residential Development at Ma Tso Lung Res 3	MTL_RES3_1	Planned Residential Development at Ma Tso Lung Res 3	R	45
MTL_RES6[2]	Planned Residential Development at Ma Tso Lung Res 6	MTL_RES6_1	Planned Residential Development at Ma Tso Lung Res 6	R	45
		MTL_RES6_2	Planned Residential Development at Ma Tso Lung Res 6	R	45
		MTL_RES6_3	Planned Residential Development at Ma Tso Lung Res 6	R	45

Notes:

(1) Residential – R; Educational Institution – E.

(2) The estimated number of storeys is based on the proposed maximum building height +155mPD and an assumption of 3-meter floor-to-floor height.

(3) As stated in the approved EIA Report for the Development of Lok Ma Chau Loop (AEIAR-176/2013), central air conditioning systems are assumed to be installed in academic and administrative buildings within the Project Area. Consequently, these buildings are not considered as noise sensitive receivers (NSRs). Only student hostels and staff quarters are identified as NSRs in this assessment. The indicative locations of NSRs TL1, TL2, and TL3 are adopted from the approved EIA Report for the Development of Lok Ma Chau Loop (AEIAR-176/2013).

Table 4.8 Representative Noise Sensitive Receivers (NSRs) and Selected Noise Assessment Points (NAPs) for Fixed Noise Assessment

NSR ID	NSR Description	NAP ID	NAP Description	Land Use(1)	Number of Storey
PH	Village Houses near Ping Hang	PH1	Temporary Structure to the South of Lamp Post VG3160	R	1

Note:
(1) Residential – R.

4.4.7 For the fixed noise impact, the ASR of a NSR is determined by the type of area where the NSR is located and the presence of any influencing factors such as major roads and industrial areas. The identified NSR (PH1) is situated in a region characterised by village-type development. Therefore, the type of area where the NSR located is classified as “Rural area, including country parks or village type developments”. Since there are no major roads or industrial areas in proximity to PH1, it is not affected by any influencing factors. As such, the ASR is classified as "A."

4.4.8 To establish the criteria for fixed noise impact assessment, a prevailing background noise measurement was conducted from 20 August 2025 to 22 August 2025. A-weighted Sound Pressure Level (SPL) of $L_{90\ (1\text{-}hr)}$ dB(A) is adopted to present the background noise measurement results. The locations of noise measurement are presented in **60722948/A10/FIGURE 4.4**. The measured prevailing noise levels and the noise criteria for fixed noise impact are summarised in **Table 4.9** and **Table 4.10**, respectively.

Table 4.9 Summary of Prevailing Background Noise Measurement

ID	Measurement Location	Measured Noise Level $L_{90(1\text{-}hr)}$, dB(A)		
		0700-1900 (Daytime)	1900-2300 (Evening time)	2300-0700 (Nighttime)
N1	Lamp Post VG3160	52	52	48

Notes:
(1) Façade correction (+3 dB(A)) has been adopted.
(2) Measurement Duration: 1 hour

Table 4.10 Area Sensitivity Rating and Noise Criteria for Fixed Noise Impact Assessment

NSR	ASR (1)	ANL-5 (Daytime & Evening Time / Nighttime) (A)	Monitoring Location	Measured Prevailing Noise Levels (Day & Evening Time / Nighttime) (B)	Criteria dB(A) min. of (A) and (B)
PH1	A	55/45	N1	52/48	52/45

Note:
(1) ASR is determined with reference to Table 1 of IND-TM.

4.5 Identification of Environmental Impacts

Construction Phase

4.5.1 The potential sources of noise impact during construction phase of the Project have been identified to be from the use of PME for the various construction activities. The major construction activities to be carried out include site clearance and formation, earthworks, slope works, construction of underpass, viaducts, at-grade roads, footpaths and retaining structures within the Project Boundary. Details of the proposed construction methods is described in **Section 2** of this EIA Report as well as construction programme presented in **Appendix 2.2**.

4.5.2 According to **Section 2**, the ECR starts to descend and become depressed road in front of ecological area (EA) in the Loop. The depressed road will be constructed by open cut excavation while the underpass will be constructed by cut-and-cover construction method. The carriageway section of the ECR near Shun Yee San Tsuen and Ma Tso Lung San Tsuen will be in form of viaduct structure which will be constructed by using precast concrete construction method. Proposed slope works for the project would include retaining walls and soil nails, which would drilling for soil nail installation and either precast or cast in-situ wall construction. Based on the best-available information during the preparation of this EIA Report, majority of the Project construction activities will be carried out during non-restricted hours (0700 to 1900 of any day not a Sunday or general holiday). No percussive piling works will be required.

4.5.3 The construction phase of the Project is anticipated to commence in Q1 / Q2 2027 and be completed by Q4 Year 2031. Referring to **Section 2.10**, potential concurrent projects within the 300m assessment area and their respective tentative construction period are listed in **Table 4.11**. The locations of the potential concurrent projects are shown in **60722948/A10/FIGURE 2.9**.

Table 4.11 List of Potential Concurrent Projects

No.	Potential Concurrent Projects	Tentative Construction Period
1	Development of the Loop – Main Works Package 1	Year 2021 – 2027
2	Development of Hong Kong-Shenzhen Innovation and Technology Park at the Loop	To be completed by 2035 [1]
3	Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Feasibility Study	Year 2028 – 2034
4	Northern Metropolis Highway – Kwu Tung Section	Commission in about 3 to 4 years after the commissioning of the San Tin Section, in or before 2036.

Note:
[1] The timeline mentioned refers to Chapter 8 of the "*Development Outline for the Hong Kong Park of the Hetao Shenzhen-Hong Kong Science and Technology Innovation Co-operation Zone*." For further details, please refer to the document - https://www.itib.gov.hk/assets/files/%E6%B2%B3%E5%A5%97%E6%B7%B1%E6%B8%AF%E7%A7%91%E6%8A%80%E5%89%B5%E6%96%B0%E5%90%88%E4%BD%9C%E5%8D%80%E9%A6%99%E6%B8%AF%E5%9C%92%E5%8D%80%E7%99%BC%E5%B1%95%E7%B6%B1%E8%A6%81%20(EN).pdf).

4.5.4 *For the Development of the Project – Main Works Package 1 and Development of Hong Kong Shenzhen Innovation and Technology Park (the Park) at the Loop*, concurrent works between these projects could be minimised at the same interfacing areas via efficient and close liaison between Contractors. Cumulative construction noise impact shall be assessed in the Construction Noise Management Plan (CNMP), which includes quantitative construction noise impact assessment, proposal of noise mitigation measures, and noise monitoring and auditing programme.

4.5.5 For the *MTL Project*, the planning study is still on-going and there is no definitive design information at the time of preparation of this EIA. Nevertheless, it is anticipated that their studies would consider all committed projects in the vicinity, including but not limited to this Project during their subsequent study. As with all government projects, it is anticipated that best practices to abate construction noise impacts will

be implemented where practicable. On this basis, adverse cumulative construction noise impacts from these projects are not anticipated.

4.5.6 *The Northern Metropolis Highway – Kwu Tung Section* (hereinafter refers to as "NMH"), which is a designated project, is to enhance the connectivity between the interchange at Fanling and the proposed interchange at Man Kam To Road to alleviate the future traffic demands generated by the future developments in the concerned locations. Based on the available information at the time of this EIA study, the project is expected to commission in about 3 to 4 years after the commissioning of the San Tin Section, in or before 2036, to alleviate the potential traffic congestion at Fanling Highway as well as to support New Development Areas. Concurrent work between the NMH and the Project may not overlap. To address this, effective coordination and communication with the NMH project proponent will be established to ensure that the cumulative construction noise levels at representative NSRs would not exceed the noise criteria.

Operation Phase

Road Traffic Noise

4.5.7 Road traffic noise impact from the road network within the 300m assessment area, including "Project roads", "Existing roads" and "Planned Roads from other Projects" will be the noise source during operation phase of the Project. The extent of "Project Roads" is presented in **60722948/A10/FIGURE 4.5**.

4.5.8 Potential concurrent projects have been identified in **Section 4.5.3**. Cumulative operation road traffic noise impact from the abovementioned projects have been considered in this study.

Fixed Noise

4.5.9 There is no existing fixed noise source identified within assessment area. A section of the Project from the Loop to Ping Hang includes an underpass featuring two ventilation openings. These openings rely solely on natural ventilation, and it has been confirmed that there will be no mechanical ventilation installed. Therefore, it is anticipated that there will be no potential fixed plant noise sources associated with these ventilation openings.

4.5.10 A pumping station is proposed as part of the Project and is listed in **Table 4.12**. The dimensions and size of the pumping station will be determined during the later design stage. An indicative location of the proposed pumping station is presented in **60722948/A10/FIGURE 4.6**. At this stage, the design of pumping station is fully enclosed by concrete structure with the exhaust louver oriented to the north, facing Hoo Hok Wai. This configuration is intended to minimise the impact of potential fixed noise sources.

Table 4.12 Fixed Noise Source from the Project

Fixed Noise Source ID	Description
PS	Pumping Station (PS)

4.6 Assessment Methodology

Construction Phase

4.6.1 In accordance with the requirements given in the Clause 1.2.2 of Appendix C of Study Brief, construction noise impact was assessed qualitatively. The general approach is summarised below:-

- Locate the representative NSRs which will most likely be affected by noise from the construction work;
- Determine the items of PME for each discrete construction activity, based on available information or agreed plant inventories;
- Identify the major noise sources and construction activities;
- Evaluate the potential impact on the NSRs qualitatively;
- Propose suitable mitigation measures such as quieter construction method and equipment, barrier, enclosure, etc; and
- Propose to submit a Construction Noise Management Plan (CNMP), which contains quantitative construction noise impact assessment, to EPD.

4.6.2 The tentative construction programme is presented in **Appendix 2.2** and the construction plant inventory, which has been confirmed by the Project Engineer of the Design Consultant to be practical and suitable for the proposed works, is presented in **Section 4.7.1**.

4.6.3 Upon identifying the major noise sources and activities, quieter construction method and equipment shall be identified to mitigate construction noise impact. Additional mitigation measures including barrier and enclosure shall be reviewed and proposed to mitigate construction noise impact.

4.6.4 The CNMP, which shall be prepared before tender invitation and before commencement of construction of the Project, will contain an updated quantitative construction noise impact assessment, quieter construction method and equipment, additional direct mitigation measures, and construction noise impact monitoring and auditing program.

Operation Phase

Road Traffic Noise

4.6.5 Road traffic noise impact assessment is predicted based on the traffic flows, following strictly the procedures stipulated in the “Calculation of Road Traffic Noise (CRTN)” (1988) published by Department of Transport, UK. Road traffic noise is presented in terms of noise levels exceeded for 10% of the one-hour period having the peak traffic flow (i.e. $L_{10,\ 1hour}$, dB(A)). A 2.5 dB(A) façade reflection and correction factors for effects due to gradient, distance, view angle, road surface and barriers would be included in the assessment.

4.6.6 Road traffic noise levels were predicted based on the maximum traffic projections within 15 years upon commencement of operation of the Project according to Appendix C of the EIA Study Brief and Annex 13 of the EIAO-TM. Based on the latest

project programme, the Project is expected to commence operation in Year 2031. The assessment year for road traffic noise was taken as Year 2046, which is the maximum traffic projection within 15 years after operation of the Project. The peak hour traffic forecast of "with Project" scenario in Year 2046 is presented in **Appendix 4.2**. Since AM peak period exhibited a higher traffic forecast on the ECR, which is the primary source of the project road, AM peak traffic data is selected in the assessment. Endorsement of the methodology of traffic forecast for this study from Transport Department and traffic consultant's confirmation on their compliance with the endorsed methodology in forecasting the traffic is presented in **Appendix 4.3**.

4.6.7 With reference to Clause 2.3.1.2 of Appendix C of Study Brief, four scenarios shall be studied. The following scenarios are studied in the assessment:-

- Scenario without project in Year 2046;
- Unmitigated scenario in Year 2046;
- Mitigated scenario in Year 2046; and
- Prevailing scenario for indirect mitigated measures eligible assessment.

4.6.8 With consideration of the scope of the Project, road sections have been classified as the following categories for the purpose of the road traffic noise assessment:-

- "Project Roads" mean roads that are completely new or a modification to an existing road section due to the proposed Project. In this assessment, the road sections including ECR and the Border Road to be altered under this Project are categorised as "Project Road"; and
- "Other Roads" mean roads that are existing roads with road alignments remain unchanged, and the planned roads from other projects. In this assessment, the existing roads (Border Road and Liu Pok Road), as well as the planned road from other projects (i.e. MTL Project, Northern Metropolis Highway and the Loop), are categorised as "Other Roads".

4.6.9 For the purpose of the road traffic noise assessment, roads within 300m from the Project boundary have been included in the assessment. Road sections included in the assessment and the extent of Project roads are presented in **60722948/A10/FIGURE 4.5**. The characteristics of the road network such as road width, surface type and traffic flow have been considered in the assessment. The standard 0.8m solid parapet as designed along all at-grade roads and the viaduct has been included in the assessment. CEDD and HyD were also consulted on the design on road networks for the planned roads from MTL Project, Northern Metropolis Highway and the Loop for road traffic noise assessment. No low noise road surfacing (LNRS), existing and committed noise mitigation measures are identified within the assessment area.

4.6.10 According to Clause 4.7.6 of EIAO Guidance Note No. 12/2023 *Road Traffic Noise Impact Assessment* (hereinafter referred to as "EIAO-GN 12/2023"), direct mitigation measures would be proposed for Project roads if adverse noise impact is predicted. If the NSRs are affected by noise from existing roads, direct mitigation measures are required to reduce the noise from Project roads to a level that it:-

- is not higher than the noise standard; and

- has no significant contribution (less than 1.0 dB(A)) to the cumulative noise level, if the overall noise level (i.e. noise from the new roads together with other existing roads) exceeds the noise standard.

4.6.11 According to Clause 4.8 of EIAO-GN 12/2023 and Annex 13 of the EIAO-TM, where the predicted noise impact exceeds the applicable noise criteria and upon exhausting of practicable noise mitigation measures stipulated in Clause 6.1, Annex 13 of the EIAO-TM, indirect technical remedies (ITR) in the form of acoustic insulation and air-conditioning is to be considered as a "last resort" to mitigate the residual noise impact. The eligibility for ITR is suggested to be tested against the following three criteria:-

- the predicted overall noise level, $L_{10(1hour)}$, from the Project roads, together with other traffic noise in the vicinity must be above a respective noise level (for example, 70 dB(A) for domestic premises and 65 dB(A) for educational institutions);

- the predicted overall noise level is at least 1.0 dB(A) more than the prevailing traffic noise level; and

- the contribution to the increase in the predicted overall noise level from the Project roads must be at least 1.0 dB(A).

Fixed Noise Source

4.6.12 In accordance with the requirements given in the Clause 3.2.3 of Appendix C of Study Brief, fixed noise impact was assessed qualitatively. The general approach is summarised below:-

- Locate the representative NSRs which will most likely be affected by noise from the fixed noise source;

- Determine the inventory and sound power level for each fixed noise source, based on available information or agreed plant inventories;

- Evaluate the potential impact on the NSRs qualitatively;

- Propose suitable mitigation measures such as using quieter fixed plant, use of silencer, installation of acoustic louvre, installation of noise barrier, and installation of noise enclosure, etc; and

- Propose to submit a Fixed Noise Management Plan (FNMP), which contains quantitative fixed noise impact assessment, to EPD for approval.

4.7 Prediction and Evaluation of Environmental Impacts

Construction Phase

4.7.1 Potential construction noise impact would arise from the use of PME during construction phase of the Project. The plant inventory which has been confirmed by the Project Engineer of the Design Consultant to be practical and suitable for the proposed works is presented in **Table 4.13**. No percussive piling work is proposed for this project.

Table 4.13 Plant Inventory for Key Construction Activities

Construction Activities	Plant Inventory	
Site Formation	• Air compressor • Dump truck • Dozer • Handheld Breaker • Long reach excavator	• Standard Excavator • Compactor • Portable electric pump • Water pump
Remove Temporary Site Formation	• Dump truck • Long reach excavator • Portable electric pump	• Standard Excavator • Water pump
Piling (Bored Pile)	• Air compressor • Crane lorry • Crawler crane • Generator • Concrete lorry mixer • Water pump • Silo tank • Oxy-acetylene set	• Mobile crane • Reverse circulation drill • Rotary table • Vibratory hammer • Bar bender • Dump Truck • Concrete pump • Excavator
Excavation and Lateral Support (ELS)	• Crane lorry • Crawler crane • Drilling rig • Excavator • Generator • Hydraulic hammer	• Oxy-acetylene set • Sediment tank • Vibratory hammer • Water pump • Welding machines
Reinforced Concrete (RC) Structure	• Crane lorry • Generator • Vibratory poker • Concrete lorry mixer • Water pump	• Mobile crane • Pump truck • Bar bender • Dump Truck • Concrete pump • Excavator
Backfill of Openings	• Compactor (roller) • Crane lorry	• Dump truck • Excavator
Pre-bored Socket H Piles	• Air compressor • Crane lorry • Crawler crane • Generator • Concrete lorry mixer • Water pump • Silo tank	• Mobile crane • Reverse circulation drill • Rotary table • Vibratory hammer • Bar bender • Dump Truck • Concrete pump

Construction Activities	Plant Inventory	
	• Oxy-acetylene set • Excavator	• Drilling rig • Concentric ring bit • Hydraulic jack
Open Cut without Lateral Support	• Dump truck	• Excavator
Ground Investigation (GI Works)	• Crane lorry • Generator	• GI drill rig • Water Pump
Lifting Frame / Segment	• Crawler crane • Generator	• Mobile crane
Slope Works	• Air compressor • Drilling rig • Excavator • Generator	• Grout Mixer • Grout Pump • Lorry
Roadworks	• Excavator • Concrete lorry mixer • Roller • Water Pump • Dump Truck	• Generator • Vibratory poker • Asphalt paver • Mobile Crane • Bar bender

4.7.2 For the work sites near Ping Hang, key construction activities include construction of portal structures and underpass sections and construction of slip roads connecting to Border Road. The distance from the nearest village houses to the Project boundary is approximately 6m as shown in **60722948/A10/FIGURE 4.2b**. Considering that the houses at Ping Hang are only 1-storey, with proper implementation of appropriate mitigation measures and good site practices (e.g. use of Quality Powered Mechanical Equipment (QPME) / quieter construction methods, site hoarding with higher surface density and height (i.e. not less than 3m), noise barriers, noise enclosures and locating mobile plant as far away as possible and practicable from NSRs, etc.), no adverse construction noise impact on the NSRs at Ping Hang is anticipated.

4.7.3 There are four 1 to 2-storey village houses at Tse Koo Hang with the nearest located at about 20m from the Project boundary (**60722948/A10/FIGURE 4.2c**). Major construction activities include slope works, site formation, construction of at-grade roads, construction of columns and decks for viaduct sections, etc. Some of those construction activities would be partly screened by natural terrain. With proper implementation of appropriate mitigation measures and good site practices (e.g. use of QPME / quieter construction methods, site hoarding with higher surface density and height (i.e. not less than 3m), noise barriers, noise enclosures and locating mobile plant as far away as possible and practicable from NSRs, etc.), no adverse construction noise impact on the NSRs at Tse Koo Hang is anticipated.

4.7.4 Key construction activities on the eastern side of the Project (**60722948/A10/FIGURE 4.2c** and **4.2d**) include slope works, site formation, construction of viaduct sections.

Existing NSRs in the vicinity include Ma Tso Lung Shun Yee San Tsuen, Ma Tso Lung San Tsuen, village houses near Ma Tso Lung Road and village houses near the junction of Ma Tso Lung Road and Liu Pok Road. There is also a planned residential development at MTL Project. Their ranges of the nearest distance to the Project boundary are approximately <5m to 115m. As shown in **60722948/A10/FIGURE 4.2d**, MTLSYST8, MTLSYST9, MTLRLPR3, MTLST4, MTLST5, MTLST6 and MTLST8 are the closest NAPs to the Project boundary (approximately <5m) with potential noise impact anticipated without the adoption of noise mitigation measures. Proper implementation of appropriate mitigation measures and good site practices (e.g. use of QPME / quieter construction methods, site hoarding with higher surface density and height (i.e. not less than 3m), noise barriers, noise enclosures and locating mobile plant as far away as possible and practicable from NSRs, etc.) would be required to minimise the associated construction noise impacts.

Operation Phase Road Traffic Noise

4.7.5 Road traffic noise impact assessment has been conducted for the representative NSRs in the Year 2046. The predicted road traffic noise levels at the representative NSRs under the unmitigated scenario are presented in **Table 4.14** while the detailed traffic noise assessment results are presented in **Appendix 4.4**.

4.7.6 Referring to **Table 4.14** and **Appendix 4.4**, the predicted cumulative road traffic noise levels at the representative NSRs are in the range of 54 dB(A) to 78 dB(A). All existing NSRs comply with the noise criteria of 70dB(A), except for MTLR5, MTLRS1 and MTLST3. Predicted road traffic noise levels at the planned NSRs at Development of Ma Tso Lung (Res 3 and Res 6) and the Loop would exceed the noise criteria of 70dB(A). Direct mitigation measures should be considered to alleviate the adverse traffic noise impact from the Project roads for those NSRs to a level that it:-

- is not higher than the noise standard; and
- has no significant contribution (less than 1.0 dB(A)) to the cumulative noise level, if the overall noise level (i.e. noise from the new roads together with other existing roads) exceeds the noise standard.

Table 4.14 Summary of Predicted Road Traffic Noise Assessment Results under Unmitigated Scenario (Year 2046)

NSR ID	NSR Description	NAP ID	NAP Description	Land Use[1]	Noise Criteria, $L_{10(1\text{-hr})}$, dB(A)	Predicted Noise Level, $L_{10(1\text{-hr})}$ dB(A)[2][3]				Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads[5]	Project Roads[6]	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria[4]	
Existing NSRs										
MTLRLPR	Village Houses near the junction of Ma Tso Lung Road and Liu Pok Road	MTLRLPR1	Temporary Structure to the East of Lamp Post N4001	R	70	63	58	65	-	N
		MTLRLPR2	Temporary Structure to the South of Lamp Post CD0749	R	70	65	57	66	-	N
		MTLRLPR3	Temporary Structure to the West of Lamp Post CD0983	R	70	66	57	67	-	N
MTLR	Village Houses near Ma Tso Lung Road	MTLR1	Temporary Structure to the Northeast of Lamp Post VA4082	R	70	59	64	65	-	N
		MTLR2	Village House to the North of Lamp Post VG3124	R	70	57	57	60	-	N
		MTLR3	Village House to the North of Lamp Post VA4080	R	70	58	62-63	63-64	-	N
		MTLR4	Temporary Structure to the North of Lamp Post VD9113	R	70	56	67	67	-	N
		MTLR5	Village House to the Northeast of Lamp Post VD9113	R	70	53-55	70-**73**	70-**73**	18.2	Y
MTLRS	Village Houses near the Southern Area of Ma	MTLRS1	Village House near to Lamp Post N3984	R	70	**72**	64	**73**	0.6	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use[1]	Noise Criteria, $L_{10(1\text{-}hr)}$, dB(A)	Predicted Noise Level, $L_{10(1\text{-}hr)}$ dB(A)[2][3]				Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads[5]	Project Roads[6]	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria[4]	
	Tso Lung Road									
MTLST	Ma Tso Lung San Tsuen	MTLST3	Temporary Structure to the North of Lamp Post VD3133	R	70	**71**	58	**71**	0.2	N
MTLSYST	Ma Tso Lung Shun Yee San Tsuen	MTLSYST1	Village House to the Northeast of Lamp Post EC1009	R	70	55	63-64	64	-	N
		MTLSYST2	1 Ma Tso Lung Shun Yee San Tsuen Block A	R	70	53	62	62	-	N
		MTLSYST3	6 Ma Tso Lung Shun Yee San Tsuen Block A	R	70	52	61	61	-	N
		MTLSYST4	2 Ma Tso Lung Shun Yee San Tsuen Block B	R	70	49	60	60	-	N
		MTLSYST5	1 Ma Tso Lung Shun Yee San Tsuen Block C	R	70	48	60	60	-	N
		MTLSYST6	4A Ma Tso Lung Shun Yee San Tsuen Block C	R	70	50-54	60-61	60-62	-	N
		MTLSYST7	1 Ma Tso Lung Shun Yee San Tsuen Block D	R	70	47	60	60	-	N
		MTLSYST8	5 Ma Tso Lung Shun Yee San Tsuen Block D	R	70	51	60	60	-	N
		MTLSYST9	Village House to the East of Lamp Post N4004	R	70	59	59-60	62-63	-	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use[1]	Noise Criteria, $L_{10(1\text{-}hr)}$, dB(A)	Predicted Noise Level, $L_{10(1\text{-}hr)}$ dB(A)[2][3]				Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads[5]	Project Roads[6]	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria[4]	
PH	Village Houses near Ping Hang	PH1	Temporary Structure to the South of Lamp Post VG3160	R	70	54	41	54	-	N
TKH	Village Houses near Tse Koo Hang	TKH1	Village House to the West of Lamp Post VG4766	R	70	39-41	62-63	62-63	-	N
		TKH2	Village House to the Southwest of Lamp Post VG4766	R	70	38-39	60-62	60-62	-	N
		TKH3	Village House in DD96 1427	R	70	38-39	63-64	63-64	-	N
		TKH4	Temporary Structure to the East of Lamp Post VG3141	R	70	43	70	70	-	N
Planned NSRs										
TL	Planned Development within the Loop	TL1	Planned Student Hostel and Staff Quarter	R [7]	70	66	56-58	67	-	N
		TL2	Planned Student Hostel and Staff Quarter	R [7]	70	**71-75**	55-57	**71-75**	0.2	N
		TL3	Planned Student Hostel and Staff Quarter	R [7]	70	**72-73**	54-55	**72-73**	0.1	N
MTL_RES3	Planned Residential Development at Ma Tso Lung Res 3	MTL_ RES3_1	Planned Residential Development at Ma Tso Lung Res 3	R	70	**71-78**	55-64	**71-78**	0.6	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use(1)	Noise Criteria, $L_{10(1\text{-}hr)}$, dB(A)	Predicted Noise Level, $L_{10(1\text{-}hr)}$ dB(A)(2)(3)				Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads(5)	Project Roads(6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
MTL_RES6	Planned Residential Development at Ma Tso Lung Res 6	MTL_RES6_1	Planned Residential Development at Ma Tso Lung Res 6	R	70	**71-72**	57-69	**72-74**	**1.8**	Y
	Planned Residential Development at Ma Tso Lung Res 6	MTL_RES6_2	Planned Residential Development at Ma Tso Lung Res 6	R	70	**72-74**	57-70	**73-74**	**1.9**	Y
		MTL_RES6_3	Planned Residential Development at Ma Tso Lung Res 6	R	70	**73-75**	56-70	**73-76**	**2.0**	Y

Notes:
(1) Residential – R;
(2) Boldfaced and underlined values indicate exceedance of relevant noise criteria.
(3) Noise levels are rounded up to nearest integer to determine the compliance of the criteria.
(4) Maximum Project Road contribution for NSRs with overall noise level exceeding relevant criteria, "-" indicates predicted overall noise level at the NSR would comply with noise criteria.
(5) Other roads include existing Road, planned road from other Projects.
(6) Project roads include the section of Border Road to be altered under this Project and ECR.
(7) As stated in the approved EIA Report for the Development of Lok Ma Chau Loop (AEIAR-176/2013), central air conditioning systems are assumed to be installed in academic and administrative buildings within the Project Area. Consequently, these buildings are not considered as noise sensitive receivers (NSRs). Only student hostels and staff quarters are identified as NSRs in this assessment. The indicative locations of NSRs TL1, TL2, and TL3 are adopted from the approved EIA Report for the Development of Lok Ma Chau Loop (AEIAR-176/2013).

Operation Phase Fixed Noise

4.7.7 Subject to the preliminary design of the pumping station, the nearest NSR at Ping Hang (i.e. NSR ID PH) located approximately 40m from the proposed site of pumping station. The noise sources, including electric water pumps, motor and generator will be installed in the pumping station are expected to be housed within the reinforced concrete building with the exhaust louver oriented to the north facing Hoo Hok Wai. Details of the associated fixed plant inventories are presented in **Table 4.15**. With proper implementation of appropriate mitigation measures, no adverse fixed noise source impact on the NSR is anticipated.

Table 4.15 Fixed Plant Inventory

Fixed Noise Source ID	Plant Inventory	No. of Plant	Sound Power Level, dB(A)	Reference (1)(2)
PS	Wet Well Pump	3	90	DSD GS 3.3.1 (e)-1
	Submersible Pump	1	85	CNP 283
	Motor (Coarse Screen)	1	88	EIA-278/2022
	Emergency Generator (3)	1	-	-
	Total SWL		96	-

Notes:
(1) The SWL of wet well pump is referenced from British Standard, code 60034-9: Rotating electrical machines. Noise limits as required under DSD general specifications 3.3.1.(e).
(2) The SWL of motor is reference from Yuen Long South Effluent Polishing Plant EIA (EIA-278/2022)
(3) The generator is used only in emergency situations and is not frequently utilised.

4.8 Mitigation of Environmental Impacts

Construction Phase

4.8.1 Although the noise mitigation effects are not easily quantifiable and the benefits may vary with site conditions and operating conditions, good site practices would be easily implemented and do not impact upon the works schedule. The good site practices listed below would be followed during construction works:-

- Only well-maintained plant would be operated on-site and would be serviced regularly during the construction period;
- Silencers or mufflers on construction equipment would be utilised and properly maintained during the construction period as practicable;
- Mobile PME would be sited as far from NSRs as possible;
- PME (such as trucks) that may be in intermittent use would be shut down between work periods or would be throttled down to minimum;
- Materials stockpiles and other structures would be effectively utilised, wherever practicable, in screening noise from on-site construction activities.

4.8.2 In order to reduce the excessive construction noise impact at the NSRs, applicable quieter construction method and PME are identified below. Further mitigation measures of movable noise barriers and temporary noise barriers are also recommended.

4.8.3 Quieter construction method such as silent piling by press-in method is adopted as an alternative of traditional sheet piling. A sheet pile is clipped and pressed under the ground. Noise can be minimised by press-in sheet piles with drilling simultaneously for piling works at harder ground. The noise impact can also be further reduced as the piling works can be completed within the shorter duration by using this method. Another quieter construction method for piling works is the large diameter bored piling method, which would replace percussive piling for piling works.

4.8.4 Other quieter construction method includes the precast concrete construction method (refer to **Section 2** for details on construction method). The precasting works will be carried out in the fabrication yard away from the site, which reduces the amount of mechanical equipment used on site such as concrete mixing trucks and therefore reducing the potential noise impact. This method can also further reduce the noise impact by allowing non-linear bridge deck construction across multiple work fronts within the shorter duration.

4.8.5 It is also feasible to adopt quieter rock or concrete breaking equipment (e.g. hydraulic rock crusher) in some activities as a mitigation measure. The Contractor should, subject to the actual site conditions, proactively adopt quieter rock breaking equipment and hydraulic crusher to replace hydraulic breaker for rock or concrete breaking to carry out the demolition / excavation works, where practicable. This method can further reduce the noise impact by percussive striking actions of breaker's chisels.

4.8.6 The Contractor(s) may be able to obtain particular models of plant that are quieter than the PMEs given in GW-TM. The use of quality PME associated with the construction works is prescribed in EPD's Quality Powered Mechanical Equipment (QPME) database, which contains the SWLs for quality/quiet PME of various types, brands and models. The Contractor should, subject to the actual site conditions, proactively adopt quality PME in all kinds of construction activities, where practicable.

4.8.7 The Contractor should select appropriate quieter construction method / equipment to minimise any potential construction noise impact to the environment. A list of quieter construction method / equipment is presented below in **Table 4.16** and should be used by the Contractor. The Contractor should explore other quieter construction method and equipment to further minimise the potential construction noise impact.

Table 4.16 Quieter Construction Method / Equipment

Construction Activities	Conventional Construction Method / Equipment	Quieter Construction Method / Equipment	Applicable to the Project (Y/N)
Site Formation	Construction Equipment • Excavator-mounted breaker • Excavator • Compactor • Handheld breaker • Air compressor	Quieter Construction Method • Use of non-explosive chemical expansion agent Quieter Construction Equipment • Use of hydraulic splitter, hydraulic crush • Quiet type saw (e.g. diamond wire saw) • Use of QPME (1)	Y
Remove Temporary Site Formation	Construction Equipment • Excavator	Quieter Construction Equipment • Use of QPME (1)	Y
Piling (Bored Pile)	Construction Method • Percussive piling Construction Equipment • Air Compressor • Generator • Mobile Crane	Quieter Construction Method • Large diameter bored piling Quieter Construction Equipment • Use of QPME (1)	Y
Excavation and Lateral Support (ELS)	Construction Method • Vibratory hammering Construction Equipment • Excavator • Generator	Quieter Construction Method • Use of silent piling such as hydraulic press-in method Quieter Construction Equipment • Use of QPME (1)	Y
Reinforced Concrete (RC) Structure	Construction Method • In-situ construction Construction Equipment	Quieter Construction Method	Y

Construction Activities	Conventional Construction Method / Equipment	Quieter Construction Method / Equipment	Applicable to the Project (Y/N)
	• Excavator • Generator • Mobile crane	• Precast concrete construction method Quieter Construction Equipment • Use of QPME (1)	
Backfill of Openings	Construction Equipment • Excavator • Compactor	Quieter Construction Equipment • Use of QPME (1)	Y
Pre-bored Socket H Piles	Construction Method • Percussive piling Construction Equipment • Air Compressor • Excavator • Generator	Quieter Construction Method • Pre-bored Socket H Piles Quieter Construction Equipment • Use of QPME (1)	Y
Open Cut Without Lateral Support	Construction Equipment • Excavator	Quieter Construction Equipment • Use of QPME (1)	Y
Ground Investigation (GI Works)	Construction Equipment • Generator	Quieter Construction Equipment • Use of QPME (1)	Y
Lifting Frame/Segment	Construction Equipment • Generator • Mobile crane	Quieter Construction Equipment • Use of QPME (1)	Y
Slopeworks	Construction Equipment • Air compressor • Generator • Mobile crane	Quieter Construction Equipment • Use of QPME (1)	Y
Road works	Construction Equipment • Vibratory Poker • Mobile crane	Quieter Construction Method • Self-compacting Concrete	Y

Construction Activities	Conventional Construction Method / Equipment	Quieter Construction Method / Equipment	Applicable to the Project (Y/N)
	• Generator • Excavator • Roller	Quieter Construction Equipment • Rubber Head Poker Vibrator • Use of QPME (1)	

Note:
(1) Quality PME available at https://www.epd.gov.hk/epd/english/environmentinhk/noise/qpme/search_gen.php?lang=eng&st=sim&valid=Y

4.8.8 The Contractor should review the applicability of quiet construction method, quieter electric construction equipment, QPME, quieter rock breaking equipment and the use of hydraulic crusher instead of rock breaking. Other available noise-reducing technologies and practices should also be considered to further minimise potential construction noise impacts. These quieter construction method and equipment shall be clearly specified in a Construction Noise Mitigation Plan (CNMP) and incorporated into the tender documents during tendering process.

4.8.9 To alleviate the construction noise impact on the affected NSRs, movable noise barriers have been proposed. Movable temporary noise barriers that can be located close to noisy plant and be moved iteratively with the plant along a worksite can be very effective for screening noise from NSRs. A typical design which has been used locally is a wooden framed barrier with a small cantilevered upper portion of superficial density no less than 14kg/m^2 on a skid footing with 25mm thick internal sound absorptive lining. The measure is particularly effective for low level zone of NSRs. A cantilevered top cover would be required to achieve screening benefits at high level of NSRs. It is anticipated that suitably designed barriers could achieve at least 5-10 dB(A) reduction.

4.8.10 The use of full enclosure has been proposed in this assessment to shelter relatively static plant including air compressor and generator. This type of enclosure is expected to provide approximately 15 dB(A) noise reduction.

4.8.11 The benefits of these techniques may vary with specific site conditions and operations. The environmental noise climate would certainly be improved with these control practices, although the improvement can only be quantified during implementation when specific site parameters are known.

4.8.12 The CNMP as detailed in EIAO-GN 9/2023 would be submitted to the Director of the EPD for agreement prior to issuance of tender and commencement of construction. The CNMP shall contain quantitative construction noise impact assessment, adopted quieter construction method and equipment, noise mitigation measures, and construction noise monitoring and auditing programme, with reference to the latest plant inventories and an implementation schedule clearly listing out the mitigation measures, the implementation party, location, and timing of implementation for inclusion in the tender document, if any. Mitigation measures recommended and requirements specified in the CNMP shall be fully implemented. Any technical constraint that would hinder the use of these quieter construction method and equipment shall be evaluated and clearly recorded in the CNMP.

Operation Phase (Road Traffic Noise)

4.8.13 With reference to Annex 13 of the EIAO-TM, where the predicted road traffic noise impacts exceed the applicable noise criteria, direct mitigation measures such as low noise road surfacing (LNRS), alternative land use, alternative siting, screening by noise tolerant buildings, acoustic windows/balconies, etc. shall be considered and evaluated in appropriate manner.

4.8.14 Referring to the predicted noise levels under unmitigated scenario as shown in **Table 4.14**, it is anticipated that overall noise levels at MTLR5, MTL_RES6_1, MTL_RES6_2, and MTL_RES6_3 have exceeded the noise criteria with significant noise contribution from the Project Road. To mitigate the impact of road traffic noise on the affected NSRs, direct at-source mitigation measures of LNRS on the "Project Road" is proposed. With these measures, the existing NSR (i.e. MTLR5) would comply with the noise criteria as stipulated in Clause 4.7.6 of EIAO-GN 12/2023. For planned NSRs at Res 6, further at-receiver mitigation measures, such as the provision of acoustic windows/balconies at the residential units in Res 6, are recommended. With the implementation of direct mitigation measures of LNRS and at-receiver mitigation measures, such as acoustic windows/balconies at the planned residential site (Res 6), all planned NSRs would comply with the criteria as stipulated in Clause 4.7.6 of EIAO-GN 12/2023.

4.8.15 Summary of mitigation measures are shown in **Table 4.17.** The predicted mitigated noise levels of all the NSRs are summarised in **Table 4.18** and presented in **Appendix 4.5.**

Table 4.17 Mitigation Measures for Road Traffic Noise Impact

ID	Type	Required At-receiver Noise Reduction, dB(A)	Approx. Length, m	NSR to be Protected / Benefited	Implementation Schedule
LNRS1	LNRS(1)	N/A	320	MTLR	Commencement of ECR
LNRS2	LNRS(1)	N/A	530	MTL_RES6	Commencement of ECR
AW1	Acoustic Windows/ Balconies (2) (3)	2.3-4.1	N/A	MTL_RES6	Population Intake of the Proposed NSR

Notes:

(1) The material proposed for the Low Noise Road Surfacing (LNRS) is "6 mm Highly Modified Stone Mastic Asphalt" (HMSMA6), in accordance with the Highways Department Guidance Notes RD/GN/011E, "Guidance Notes on Low Noise Road Surfacing on Local Roads." The final material selection will be subject to any future detail design / updates to the relevant guidance or specifications.

(2) Details refer to "Innovative Noise Mitigation Designs and Measures" on EPD's Website. (https://www.epd.gov.hk/epd/Innovative/greeny/eng/index.html)

(3) Acoustic windows/ balconies will be provided if the Project Road has significant contributions (i.e. ≥ 1.0 dB(A)) after applying LNRS.

4.8.16 The proposed at-receiver mitigation measures i.e. the provision of acoustic windows / acoustic balconies for planned residential site (i.e. Res 6) demonstrated the feasibility of implementing direct mitigation measures at-receiver end based on the best available information at the time of this EIA Report without significant implication to the future development parameters and building layout of Res 6. These mitigation measures are indicative only and are subject to further study by future developer/ project proponent, in conjunction with their detailed design. To ensure the effectiveness of the direct mitigation measures, a detailed Noise Impact Assessment (NIA) or Environmental Assessment Study (EAS) for the planned residential site shall be undertaken by the future developer / project proponent during detailed design stage. The NIA or EAS should be conducted in accordance with Chapter 9 of the HKPSG, and explore other direct at-receiver mitigation measures, such as further setbacks, extended podium, optimised building orientation, architectural features, special building design, together with the recommended mitigation measures (i.e. acoustic windows/acoustic balconies), to ensure compliance with relevant criteria based on their building layouts. As agreed with the project proponent of MTL Project, CEDD will make requests to relevant authorities for incorporating the requirements of NIA or EAS in future housing development, through land lease mechanism or otherwise, where appropriate. The detailed agreement, as confirmed through email correspondence, is provided in **Appendix 4.6**.

Table 4.18 Summary of Predicted Road Traffic Noise Assessment Results under the Mitigated Scenario (with implementation of Acoustic Window / Acoustic Balconies and Low Noise Road Surfacing)

NSR ID	NSR Description	NAP ID	NAP Description	Land Use (1)	Noise Criteria, $L_{10(1\text{-hr})}$, dB(A)	Predicted Noise Level after mitigated with LNRS, $L_{10(1\text{-hr})}$ dB(A)(2)(3)				Predicted Noise Level after mitigated with LNRS and Acoustic Windows / Balconies, $L_{10(1\text{-hr})}$ dB(A)			Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads (5)	Project Roads (6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	Min. Required Noise Reduction of Acoustic Window / Balconies, dB(A)(7)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
Existing NSRs													
MTLR LPR	Village Houses near the junction of Ma Tso Lung Road and Liu Pok Road	MTLRLPR1	Temporary Structure to the East of Lamp Post N4001	R	70	63	56	64	-	-	-	-	N
		MTLRLPR2	Temporary Structure to the South of Lamp Post CD0749	R	70	65	55	65	-	-	-	-	N
		MTLRLPR3	Temporary Structure to the West of Lamp Post CD0983	R	70	66	55	66	-	-	-	-	N
MTLR	Village Houses near Ma Tso Lung Road	MTLR1	Temporary Structure to the Northeast of Lamp Post VA4082	R	70	59	62	64	-	-	-	-	N
		MTLR2	Village House to the North of Lamp Post VG3124	R	70	57	56	60	-	-	-	-	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use (1)	Noise Criteria, $L_{10(1\text{-hr})}$, dB(A)	Predicted Noise Level after mitigated with LNRS, $L_{10(1\text{-hr})}$ dB(A)(2)(3)				Predicted Noise Level after mitigated with LNRS and Acoustic Windows / Balconies, $L_{10(1\text{-hr})}$ dB(A)			Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads (5)	Project Roads (6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	Min. Required Noise Reduction of Acoustic Window / Balconies, dB(A)(7)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
		MTLR3	Village House to the North of Lamp Post VA4080	R	70	58	60-61	62-63	-	-	-	-	N
		MTLR4	Temporary Structure to the North of Lamp Post VD9113	R	70	56	65	65	-	-	-	-	N
		MTLR5	Village House to the Northeast of Lamp Post VD9113	R	70	53-55	68-70	68-70	-	-	-	-	N
MTLRS	Village Houses near the Southern Area of Ma Tso Lung Road	MTLRS1	Village House near to Lamp Post N3984	R	70	**72**	62	**72**(9)	0.4	-	-	-	N
MTLST	Ma Tso Lung San Tsuen	MTLST3	Temporary Structure to the North of Lamp Post VD3133	R	70	**71**	55	**71**(9)	0.2	-	-	-	N
MTLSYST		MTLSYST1	Village House to the Northeast of	R	70	55	61-62	62	-	-	-	-	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use (1)	Noise Criteria, $L_{10(1\text{-hr})}$, dB(A)	Predicted Noise Level after mitigated with LNRS, $L_{10(1\text{-hr})}$ dB(A)(2)(3)				Predicted Noise Level after mitigated with LNRS and Acoustic Windows / Balconies, $L_{10(1\text{-hr})}$ dB(A)			Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads (5)	Project Roads (6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	Min. Required Noise Reduction of Acoustic Window / Balconies, dB(A)(7)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
	Ma Tso Lung Shun Yee San Tsuen		Lamp Post EC1009										
		MTLS YST2	1 Ma Tso Lung Shun Yee San Tsuen Block A	R	70	53	61	62	-	-	-	-	N
		MTLS YST3	6 Ma Tso Lung Shun Yee San Tsuen Block A	R	70	52	60	61	-	-	-	-	N
		MTLS YST4	2 Ma Tso Lung Shun Yee San Tsuen Block B	R	70	49	59	60	-	-	-	-	N
		MTLS YST5	1 Ma Tso Lung Shun Yee San Tsuen Block C	R	70	48	59	60	-	-	-	-	N
		MTLS YST6	4A Ma Tso Lung Shun Yee San Tsuen Block C	R	70	50-54	59-60	59-61	-	-	-	-	N
		MTLS YST7	1 Ma Tso Lung Shun Yee San Tsuen Block D	R	70	47	59	59	-	-	-	-	N
		MTLS YST8	5 Ma Tso Lung Shun Yee San Tsuen Block D	R	70	51	58	59	-	-	-	-	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use (1)	Noise Criteria, $L_{10(1\text{-hr})}$, dB(A)	Predicted Noise Level after mitigated with LNRS, $L_{10(1\text{-hr})}$ dB(A)(2)(3)				Predicted Noise Level after mitigated with LNRS and Acoustic Windows / Balconies, $L_{10(1\text{-hr})}$ dB(A)			Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads (5)	Project Roads (6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	Min. Required Noise Reduction of Acoustic Window / Balconies, dB(A)(7)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
		MTLS YST9	Village House to the East of Lamp Post N4004	R	70	59	57-58	61-62	-	-	-	-	N
PH	Village Houses near Ping Hang	PH1	Temporary Structure to the South of Lamp Post VG3160	R	70	54	41	54	-	-	-	-	N
TKH	Village Houses near Tse Koo Hang	TKH1	Village House to the West of Lamp Post VG4766	R	70	39-41	62	62-63	-	-	-	-	N
		TKH2	Village House to the Southwest of Lamp Post VG4766	R	70	38-39	60-62	60-62	-	-	-	-	N
		TKH3	Village House in DD96 1427	R	70	38-39	63-64	63-64	-	-	-	-	N
		TKH4	Temporary Structure to the East of Lamp Post VG3141	R	70	43	70	70	-	-	-	-	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use (1)	Noise Criteria, $L_{10(1\text{-}hr)}$, dB(A)	Predicted Noise Level after mitigated with LNRS, $L_{10(1\text{-}hr)}$ dB(A)(2)(3)				Predicted Noise Level after mitigated with LNRS and Acoustic Windows / Balconies, $L_{10(1\text{-}hr)}$ dB(A)			Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads (5)	Project Roads (6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	Min. Required Noise Reduction of Acoustic Window / Balconies, dB(A)(7)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
Planned NSRs													
TL	Planned Development within the Loop	TL1	Planned Student Hostel and Staff Quarter	R (8)	70	66	56-58	67	-	-	-	-	N
		TL2	Planned Student Hostel and Staff Quarter	R (8)	70	**71-75**	55-57	**71-75**(9)	0.2	-	-	-	N
		TL3	Planned Student Hostel and Staff Quarter	R (8)	70	**72-73**	54-55	**72-73**(9)	0.1	-	-	-	N
MTL_RES3	Planned Residential Development at Ma Tso Lung Res 3	MTL_RES3_1	Planned Residential Development at Ma Tso Lung Res 3	R	70	**71-78**	53-61	**71-78**(9)	0.3	-	-	-	N
MTL_RES6	Planned Residential Development at Ma Tso Lung Res 6	MTL_RES6_1	Planned Residential Development at Ma Tso Lung Res 6	R	70	**71-72**	55-67	**72-73**	**1.1**	2.3-3.1	70-**73**(9)	0.9	N

NSR ID	NSR Description	NAP ID	NAP Description	Land Use (1)	Noise Criteria, $L_{10(1\text{-}hr)}$, dB(A)	Predicted Noise Level after mitigated with LNRS, $L_{10(1\text{-}hr)}$ dB(A)(2)(3)				Predicted Noise Level after mitigated with LNRS and Acoustic Windows / Balconies, $L_{10(1\text{-}hr)}$ dB(A)			Whether Noise Mitigation Measures on "Project Roads" are Required (Y/N)
						Other Roads (5)	Project Roads (6)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	Min. Required Noise Reduction of Acoustic Window / Balconies, dB(A)(7)	Overall	Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria(4)	
		MTL_RES6_2	Planned Residential Development at Ma Tso Lung Res 6	R	70	**72-74**	55-67	**73-74**	**1.2**	3.6	70-**74**(9)	0.9	N
		MTL_RES6_3	Planned Residential Development at Ma Tso Lung Res 6	R	70	**73-75**	54-68	**73-75**	**1.2**	3.8-4.1	70-**75**(9)	0.9	N

Notes:

(1) Residential – R;
(2) Boldfaced and underlined values indicate exceedance of relevant noise criteria.
(3) Noise levels are rounded up to nearest integer to determine the compliance of the criteria.
(4) Maximum Project Road contribution for NSRs with overall noise level exceeding relevant criteria, "-" indicates predicted overall noise level at the NSR would comply with noise criteria.
(5) Other roads include existing Road, planned road from other Projects.
(6) Project roads include the section of Border Road to be altered under this Project and ECR.
(7) Acoustic windows/ balconies will be provided if the Project Road has significant contributions (i.e. Max. Project Roads Contribution when Overall Noise Level Exceeds Criteria ≥ 1.0dB(A)) after applying LNRS.
(8) As stated in the approved EIA Report for the Development of Lok Ma Chau Loop (AEIAR-176/2013), central air conditioning is assumed to be provided for academic and administrative buildings. Therefore, only the student hostels and staff quarters are considered NSRs.
(9) For these NSRs, which are predicted to exceed the noise criteria stipulated in Annex 5 of EIAO-TM despite having insignificant contributions from the Project Road (i.e. Project Road Contribution less than 1.0 dB(A)), the implementing appropriate mitigation measures shall be further determined by the Project Proponent(s) of the associated planned projects of NMH, MTL Project and/or by the future developers of Residential Sites Res 3 and Res 6, as well as the Loop.

4.8.17 With the implementation of the recommended direct noise mitigation measures, including LNRS, no adverse road traffic noise impact due to proposed "Project Roads" on existing NSRs near the Project is anticipated. Therefore, Prevailing Scenario and Scenario without project in Year 2046 are considered not necessary to be conducted.

4.8.18 In accordance with Clause 2.4.1(b) and Clause 2.3.2(c) of Appendix C of the EIA Study Brief, it is required to provide the estimated total number of dwellings and other NSRs that will be benefitted and protected as well as the estimated total number of dwellings, classrooms and other NSRs that will be exposed to noise impact exceeding the respective. With implementation of mitigation measures, the estimated total number of existing residential dwelling that would be exposed, benefited and protected respective to noise criteria are presented in **Table 4.19**.

Table 4.19 Estimated Number of Existing Residential Dwellings/ Rooms Benefited and Protected by Mitigation Measures

NSR Type	No. of Existing Dwellings (4)			
	Exposed to Exceedance (1)		Benefited (2)	Protected (3)
	Unmitigated	Mitigated	Mitigated	Mitigated
Residential	3	2	1	1

Notes:
(1) The dwellings would be considered as "exposed to exceedance" if the predicted noise level exceeded the noise criteria under the respective scenario.
(2) The dwellings would be considered as "benefited" if (a) the predicted noise level at the dwellings exceeded the criteria under unmitigated scenario and (b) more than 1.0 dB (A) noise reduction is predicted as a result of the applied mitigated measures.
(3) The dwellings or classroom would be considered as "protected" if (a) the predicted noise level at the dwellings or classroom exceeded the criteria under unmitigated scenario and (b) the predicted noise level at the dwellings or classroom complied with the noise criteria as a result of the applied mitigated measures.
(4) The building layout and the number of flats per storey for the residential site of the Loop and MTL Project are currently unavailable at the time of this EIA Report. Therefore, relevant information from the Loop and MTL Project has been excluded from the estimation.

Operation Phase (Fixed Noise)

4.8.19 In view of the potential fixed noise impact arising from the proposed pumping station, direct mitigation measures should be considered at source to minimise the potential fixed noise impact. Direct mitigation measures include using quieter fixed plant, use of silencer, installation of acoustic louvre, installation of noise barrier, and installation of noise enclosure, etc. would be recommended to minimise the potential fixed noise sources impact from the proposed fixed noise sources, such that the fixed noise level would comply with the criteria at the NSR.

4.8.20 A Fixed Noise Sources Management Plan (FNMP) as detailed in EIAO-GN 16/2023 would be submitted to Director of the EPD for agreement prior to issuance of tender and commencement of construction of the proposed fixed noise sources. The FNMP shall contain quantitative fixed noise source impact assessment with the latest list of fixed noise source plant inventory, recommended noise mitigation measures, commissioning test requirements and environmental monitoring and audit programme. FNMP shall include an implementation schedule clearly listing out the mitigation measures, the implementation party, location, and timing of implementation. Mitigation measures recommended and requirements specified in the FNMP shall be fully implemented.

4.9 Evaluation of Residual Impact

Construction Phase

4.9.1 With the implementation of the proposed mitigation measures, construction noise impact to the NSRs could be effectively controlled and minimised. No residual impact would be anticipated.

Road Traffic Noise

4.9.2 With the implementation of the recommended noise mitigation measures, including LNRS and at-receiver mitigation measures for the planned NSRs, such as acoustic windows/balconies, the noise levels at representative NSRs within the assessment area would comply with the traffic noise criteria. As such, residual road traffic noise impact due to the Project is not anticipated.

Fixed Noise Source

4.9.3 No residual impact would be expected with proper implementation of the recommended mitigation measures.

4.10 Environmental Monitoring and Audit

Construction Phase

4.10.1 An EM&A programme during construction phase is recommended to be established. The CNMP should evaluate the potential construction noise impacts based on the latest construction programme and construction plant inventory, and to assess the effectiveness and practicality of all proposed noise mitigation measures. The CNMP would be prepared during the design / tendering and implementation stage of the construction works. The implementation of the mitigation measures recommended in CNMP should also be audited as part of the EM&A programme. Details of the construction noise monitoring and audit programme would be provided in the CNMP .

Operation Phase

4.10.2 No adverse traffic noise impact is anticipated from Project contribution with the proposed mitigation measures in place. Road traffic noise levels should be monitored at representative existing NSR, which are in the vicinity of the recommended direct mitigation measures, during the first year after road opening. For the planned residential development (Res 6), noise monitoring should be carried out during the first year of population intake. The purpose of the monitoring is to ascertain that the recommended mitigation measures are effective in reducing the noise levels.

4.10.3 FNMP should be prepared to assess quantitatively the potential fixed noise sources impacts and to assess the effectiveness and practicality of all proposed noise mitigation measures. The FNMPs should also contain fixed noise sources commissioning test plans and monitoring and audit programme.

4.11 Conclusion

Construction Phase

4.11.1 The assessment for the potential construction noise impact of the Project has been qualitatively conducted. The assessment results indicate that, with the adoption of

proposed mitigation measures, including good site practices, the use of quieter construction method and equipment, QPME, site hoarding with higher surface density and height, deployment of construction noise barriers and enclosure etc., adverse construction noise impact to the NSRs could be effectively controlled and minimised. A CNMP, which to verify the inventory of noise sources, to assess the construction noise impact and effectiveness and practicality of all identified measures, would be prepared during the design / tendering and implementation stage of the construction works. Regular noise monitoring and site environmental audit during construction phase is recommended to ensure proper implementation of mitigation measures and good site practices.

Road Traffic Noise

4.11.2 Road traffic noise impact assessment has been conducted. The assessment results indicate that the predicted road traffic noise levels at some of the representative NSRs would exceed the noise criteria under the unmitigated scenario. With the implementation of direct noise mitigation measures of LNRS, and at-receiver mitigation measures for planned NSRs, such as acoustic windows/balconies, the noise levels at all NSRs within the assessment area would comply with the traffic noise criteria stipulated in Clause 4.7.6 of EIAO-GN 12/2023.

Fixed Noise Source

4.11.3 The assessment for the potential fixed noise impact of the Project has been qualitatively conducted. Noise impact from planned fixed plant could be effectively mitigated by implementing noise control measure at source during the detailed design stage. With the adoption of proposed mitigation measures, include using quieter fixed plant, use of silencer, installation of acoustic louvre, installation of noise barrier, and installation of noise enclosure, etc, adverse fixed noise impact to the NSRs could be effectively controlled and minimised. A FNMP, which to verify the inventory of noise sources, to assess the fixed noise impact, the effectiveness and practicality of all identified measures, would be submitted before construction of the Project. Therefore, no adverse impact from the proposed fixed noise sources of the Project would be anticipated.